



on Electronics

DOI:10.1587/transele.2024ECP5057

Publicized:2025/02/18

This article has been accepted and published on J-STAGE in advance of copyediting. Content is final as presented.

A PUBLICATION OF THE ELECTRONICS SOCIETY



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Topology Optimization of Microstrip Lines Using Twin Deep Neural Networks for Performance Prediction and Accuracy Evaluation

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SUMMARY This paper presents topology optimization of microstrip lines using twin deep neural networks (DNNs) for prediction of scattering parameters and its accuracy evaluation. Topology optimization can be accelerated by using a DNN that acts as a surrogate model for time-consuming EM simulations. However, if the prediction accuracy of the DNN for performance prediction is not high enough, the optimization will fail due to misleading caused by prediction errors. To reduce the risk of optimization failure, the present method introduced an additional DNN to evaluate the accuracy of the performance prediction. The proposed method is shown to be effective in avoiding misleading and speeding up the optimization process through numerical and experimental results.

key words: Bandpass filters, Deep learning, Microstrip lines, Surrogate model, Topology optimization.

1. Introduction

Microstrip lines (MSLs) are one of the most important components in high-frequency circuits and are widely used in a variety of high-frequency devices, such as cellular phone base stations and radars.

Topology optimization (TO) searches for optimal structure by freely deforming the device, allowing generation and annihilation of holes, unlike conventional parameter optimization (PO). Therefore, TO allows us to find novel structures with high performance. There are two approaches to TO: deterministic methods based on gradient computations including density method [1-6], level-set method [1, 7, 8], and function-expansion-based methods [9, 10], and stochastic methods based on metaheuristic algorithms such as genetic algorithm and covariance matrix adaptation evolution strategy (CMA-ES) [11]. In this paper, we focus on the stochastic TO using the normalized Gaussian network (NGnet) method, which expresses the material distribution by superposition of Gaussian functions [12, 13]. The NGnet method has advantages over the deterministic methods such as no need for gradient computations and easy application to multi-objective problems and robust optimization and also the ability to obtain smooth material boundaries. It has been shown to be effective in the design of electrical devices such as permanent magnet motors [12, 14, 15], wireless power transfer devices [16], and microwave devices [17-19].

Another stochastic method is the geometry projection method, where shapes are represented by a combination of simple rectangular or elliptical shapes [20, 21]. Note that the geometry projection method was originally proposed as a deterministic method [22]. For pixel pattern optimization which is different from the NGnet-based topology optimization presented here, the direct-binary-search (DBS) method with Bayesian optimization has been shown to be effective [23].

Although the stochastic TO methods including the NGnet method have many advantages as mentioned above, it is computationally expensive due to the population-based stochastic search. To circumvent this problem, it has been shown to be effective to use deep learning as a surrogate model for time-consuming electromagnetic (EM) simulations [24-28]. Because the deep neural network (DNN) provides the device characteristics much faster than EM simulations, it can effectively accelerate TO. For successful use of the surrogate models, they must provide sufficiently accurate predictions. However, it has been pointed out that it is difficult to accurately predict the characteristics of MSL circuits when they change significantly with minute changes in shape [29]. An example of the drastic change in the characteristics of the MSL circuit is shown in Fig. 1, where (a) and (b) show two very similar shapes, and (c) shows their frequency characteristics of scattering parameters (S-parameters). Although these two shapes are nearly identical, with differences in the connectivity in the conductors, their characteristics are quite different, making it difficult to make accurate predictions with a surrogate model. Unlike in PO, we often encounter this high sensitivity of the characteristics to the shape and topology in TO. Therefore, building a surrogate model with high accuracy would require a huge amount of training data and computational cost. Moreover, if a surrogate model with insufficient accuracy is used for optimization, misleading of the population to the wrong directions during the optimization process due to prediction errors can lead to unreliable optimization solutions.

The goal of this work is to develop a method that can perform fast optimization using surrogate models while avoiding misleading. To achieve this, we propose an optimization framework with twin DNNs. The first DNN

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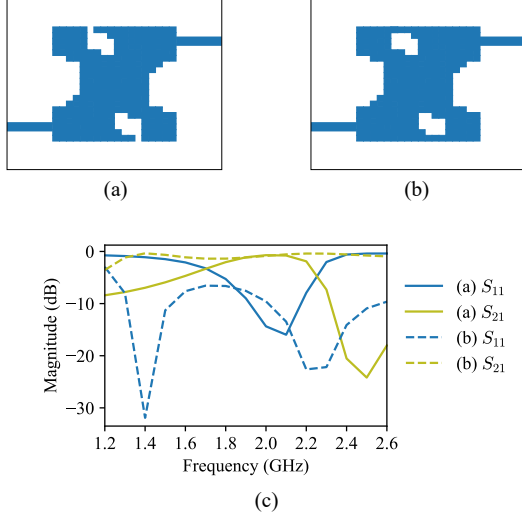


Fig. 1 Example of drastic change in characteristics due to minute changes in shape. (a) The original shape, (b) its variant with minute changes, and (c) their S-parameters. Blue and white regions represent conductor and substrate, respectively.

predicts the S-parameters of the microwave filter, and the second DNN predicts the accuracy of the first DNN. From the output of the second DNN, we determine whether or not to perform EM simulations, here finite element method, to accurately evaluate the S-parameters during optimization. DNN-based design methods for microwave devices have also been proposed [30, 31], which use a DNN as a surrogate model to speed up optimization. A highly accurate DNN with a huge data generation process (more than 300k data) and transfer learning have been presented in [30], while EM simulations are performed only on shapes for which the DNN predicts good performance in the method proposed in [31]. An approach similar to that in [31] has been applied to the topology optimization of electric motors in [24], which is the first report on accelerating topology optimization with DNN. On the other hand, the proposed method includes two DNNs and it performs EM simulation on shapes for which the second DNN predicts poor accuracy.

This paper is organized as follows. Sections 2 and 3 describe the twin DNNs and the optimization method based on it, respectively. Section 4 presents an example of microstrip bandpass filter (MSBPF) design using the proposed method along with measurement results, followed by a conclusion.

2. Twin DNNs for Prediction and Error Evaluation

To predict the frequency dependence of the S-parameters of an MSBPF, we use the first DNN shown in the upper part of Fig. 2, which takes as input the binary image of the MSBPF conductor and as output the corresponding frequency characteristics of the S-parameters. In addition, we introduce the second DNN shown in the lower part of Fig. 2. Although the input image is the same as the first DNN, the output is the root mean square error (RMSE) of the

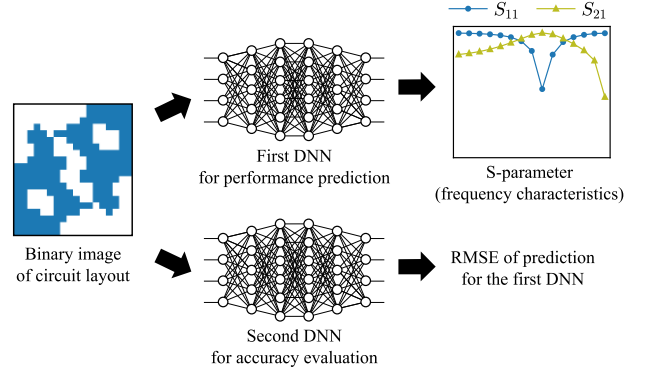


Fig. 2 Twin DNNs for prediction and error evaluation.

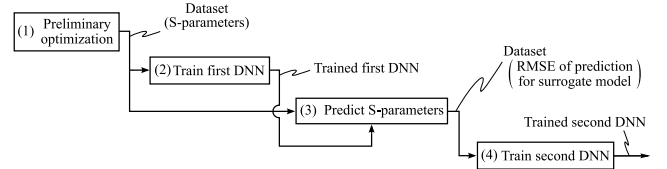


Fig. 3 Procedure for building the first and second DNNs.

prediction by the first DNN. For any input image of MSBPF, the first DNN predicts the corresponding S-parameters while the second DNN evaluates the accuracy of the prediction. These results are effectively used in the optimization process as mentioned in the next section. Although we consider the TO of MSBPFs with DNNs, the proposed method can be applied to other TOs based on any surrogate models. To implement DNNs, we use here the convolutional NN (CNN). The procedure for building the first and second DNNs is summarized in Fig. 3.

3. Topology Optimization Based on Twin DNNs

Fig. 4 illustrates the schematic diagrams of the conventional and proposed optimization methods. Here, the superscript g and subscript i denote the i -th individual of the g -th generation, $\hat{L}_i^g$ is the evaluated error by the second DNN, L_{th} is the threshold, $\hat{F}_i^g$ is the predicted value of objective function by the first DNN, and $\hat{\mathbf{F}}^g = \{\hat{F}_1^g, \hat{F}_2^g, \dots\}$. The proposed method is based on an evolutionary algorithm, where the optimization proceeds by iteratively generating (updating) and evaluating the population as shown on the left part of Fig. 4. The proposed method is characterized by the performance evaluation process shown on the right part of Fig. 4. In the conventional method (i), all individuals are evaluated by EM simulations, and in the conventional method (ii), by a surrogate model. On the other hand, the proposed method evaluates individuals by switching between the EM simulations and the surrogate model according to $\hat{L}_i^g$. We consider two methods for the proposed TO; in (i) the EM simulation is performed if $\hat{L}_i^g > L_{th}$, while in (ii), it is limited to individuals that satisfy $\hat{L}_i^g > L_{th}$ and also have good performance, thereby avoiding misleading and speeding up the process.

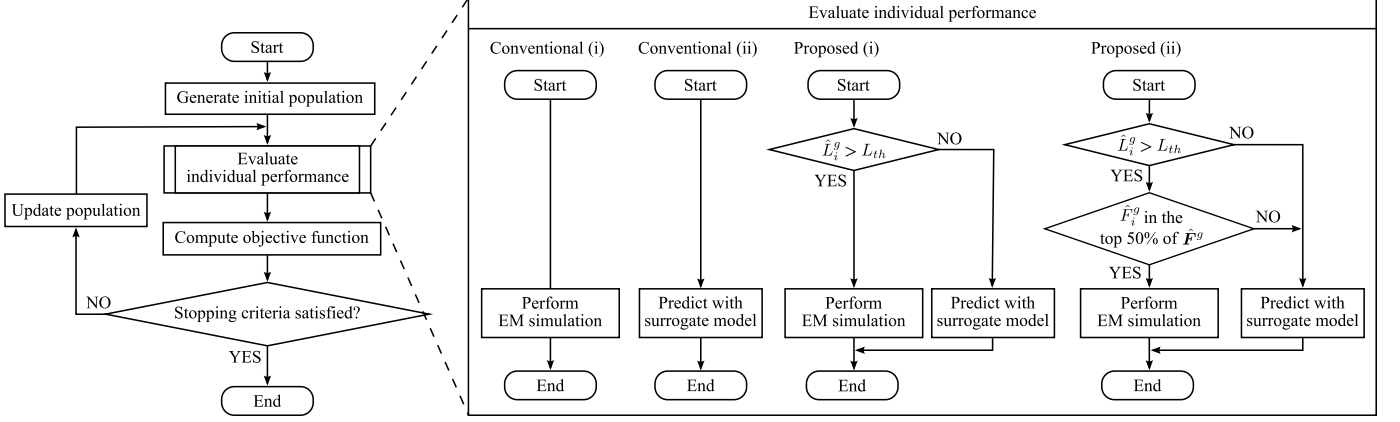


Fig. 4 Schematic diagrams of conventional and proposed optimization methods. The left part shows the optimization flow, and the right part details the individual performance evaluation.

4. TO of Microstrip Bandpass Filter

4.1 Preliminary Optimization

This section describes the preliminary optimization to generate the dataset for training the DNNs. In the NGnet method, the material distribution M_e is determined by the following equation:

$$M_e = \begin{cases} \text{Conductor,} & y(\mathbf{x}_e, \mathbf{w}) \geq 0 \\ \text{Air,} & y(\mathbf{x}_e, \mathbf{w}) < 0 \end{cases} \quad (1)$$

where $\mathbf{x}_e$ is the center position of cell e in the design region, and y is the shape function defined as

$$y(\mathbf{x}, \mathbf{w}) = \sum_{i=1}^N w_i b_i(\mathbf{x}) \quad (2)$$

$$b_i(\mathbf{x}) = G_i(\mathbf{x}) / \sum_{k=1}^N G_k(\mathbf{x}) \quad (3)$$

$$G_k(\mathbf{x}) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2} |\mathbf{x} - \mathbf{x}_i|^2\right\} \quad (4)$$

where N and $\mathbf{w} = [w_1, w_2, \dots, w_N]$ denote the number of Gaussian basis functions and weighting coefficients, respectively. G_k , σ , and $\mathbf{x}_i$ are the Gaussian basis function, its standard deviation, and its center position, respectively. The optimization problem in (5) is solved by applying TO with the NGnet method to the MSBPF shown in Fig. 5 [19].

$$F(\mathbf{w}) = \sum_{f \in \text{passband}} |S_{21}(f)|^2 + \sum_{f \in \text{stopband}} |S_{11}(f)|^2 \rightarrow \min. \quad (5)$$

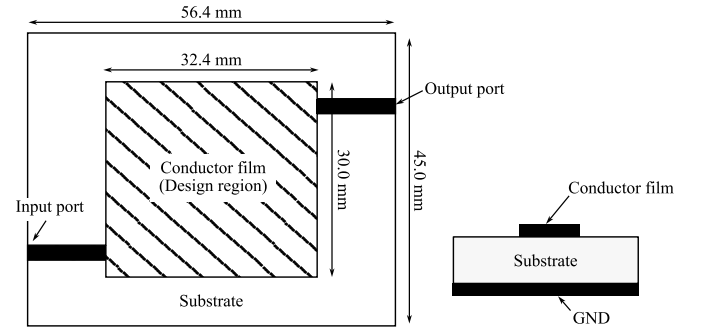


Fig. 5 Top view and cross section of the optimization model [19]. The thickness, relative permittivity, and loss tangent of the substrate are $t = 1.2$ mm, $\epsilon_r = 4.3$, and $\tan \delta = 0.011$, respectively.

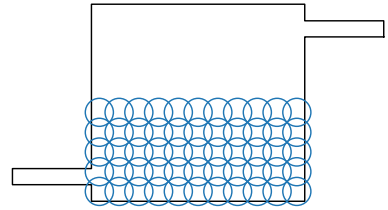


Fig. 6 Distribution of basis functions for NGnet. Blue circles represent the standard deviation of Gaussian basis functions.

where S_{11} and S_{21} are the magnitude of the S-parameters in decibels, and f is the sampling frequency. The center frequency is 2.0 GHz, and the fractional bandwidth is 20%. The distribution of the Gaussian basis and optimization settings are shown in Fig. 6 and Table 1, respectively. A constraint of 1/2 symmetry is imposed on the design region, and half of the region is optimized. The CMA-ES is used to find $\mathbf{w}$ such that F is minimized [11]. Approximately 70,000 data were generated over about 24 days by performing the same optimization many times with different random number seeds using Intel® Xeon® W-2135 CPU at 3.70GHz. All the EM simulations in this work were performed using Femtet®, a commercial computer-aided design software. From our experience, the MSBPFs whose ports are not connected by conductors in the design region do not have the desired characteristics. For this reason, we

excluded such MSBPFs, resulting in a final number of 66,973 data. Fig. 7 shows examples of the training data. Since the training data includes data in the process of optimization, some of the data are far from the desired characteristics. Note that the proposed method is expected to solve optimization problems with objective functions and constraints that are different from those in the preliminary optimization mentioned above, as will be described in section 4.4.

4.2 DNN for Prediction of S-parameters

To predict the frequency characteristics of the S-parameters, the first DNN with the architecture shown in Fig. 8 was trained with the settings shown in Table 2. The architecture of the DNNs was determined with reference to VGG [32], a simple convolutional network developed for image recognition. Different configurations of number of layers, number of nodes, dropouts, and batch normalization were found not to improve the result. The possibility of architectures with many more parameters or state of the art such as Vision Transformer [33] to improve the result is left for our future study. Adam [34] and mean squared error (MSE) defined as (6) were used as the optimizer and the loss function, respectively.

$$\text{MSELoss}(\mathbf{y}, \mathbf{t}) = \frac{1}{M} \sum_{m=1}^M (y_m - t_m)^2 \quad (6)$$

where $\mathbf{y} = [y_1, \dots, y_M]$ and $\mathbf{t} = [t_1, \dots, t_M]$ represent the predicted and true values, respectively, and M represents the number of training data.

The best model parameters when the minimum validation loss was not updated for 100 consecutive epochs were adopted. The loss curves are shown in Fig. 9. It took 73 minutes for training with Intel® Core™ i9-10940X CPU

at 3.30GHz and NVIDIA RTX™ A5000 with a minimum validation loss of 0.26. Fig. 10 shows the prediction results on the test data. This result shows that the S-parameters in 1.9-2.1 GHz for S_{11} and 2.5-2.6 GHz for S_{21} are particularly sensitive to shape, making them difficult to predict accurately and leading to large prediction errors. This is due to the frequency dependence of the variance of the dataset generated by the preliminary optimization. Because the variance of S_{11} around the passband and that of S_{21} around the stopband tend to be large, their prediction is relatively difficult. The highest correlation coefficient between ground truth and predicted values was 0.94 for S_{11} at 2.0 GHz and the lowest was 0.75 for S_{21} at 1.6 GHz.

Table 1 Settings for preliminary optimization.

Number of design variables	55
Optimization algorithm	CMA-ES
Number of generations	80
Population size λ	16
Initial value of distribution mean m	0
Initial value of step-size σ	0.3

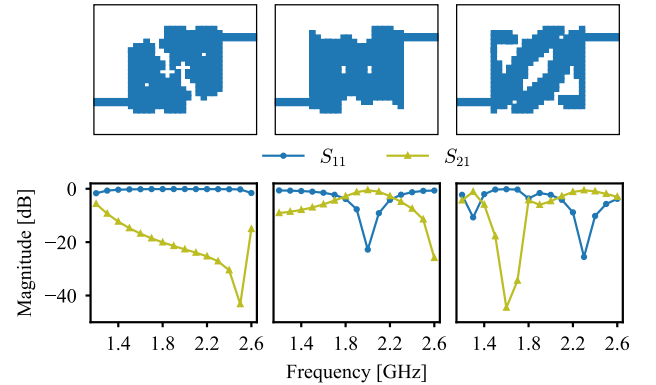


Fig. 7 Examples of training data obtained by preliminary optimization. The upper part shows the shapes of the MSBPFs, and the lower part shows the corresponding frequency characteristics of the S-parameters.

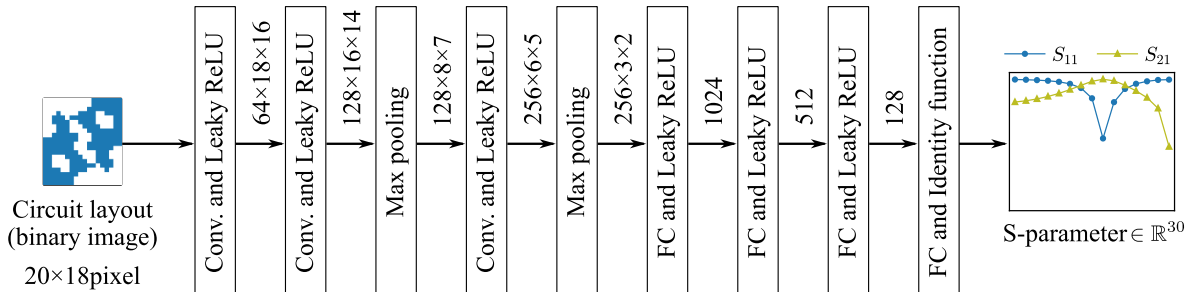


Fig. 8 Architecture of the first DNN for prediction of S-parameters. Conv. and FC represent the convolutional and fully connected layers, respectively. Angle of the negative slope for Leaky ReLU function is 0.01. The output data, S-parameters, are standardized.

Table 2 Training settings for the first DNN.

Dataset size	66,973
Data split ratio (training : validation : test)	7:1:2
Update algorithm	Adam
Loss function	MSE (6)
Batch size	128
Initial learning rate	0.001
Number of epochs	140

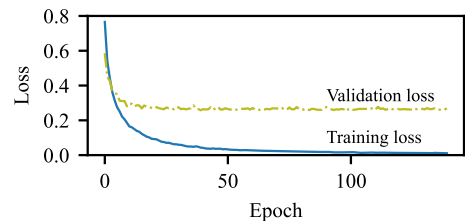


Fig. 9 Loss curves for the first DNN.

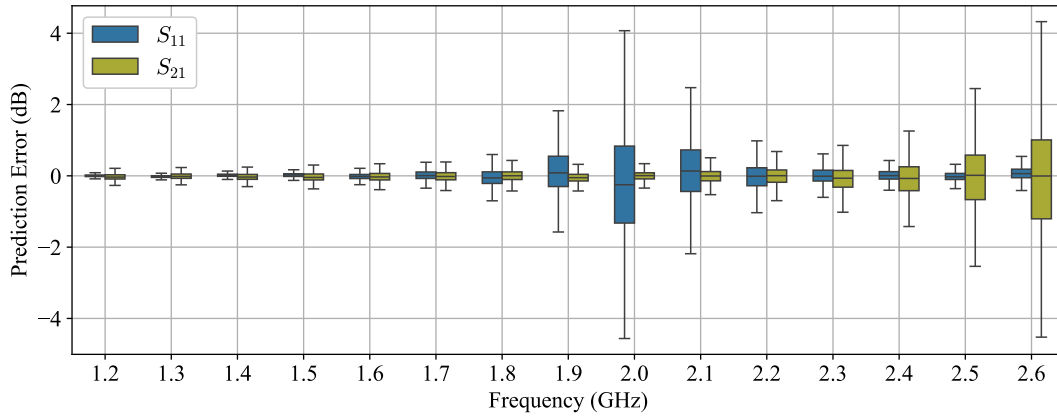


Fig. 10 Prediction results on the test data with the first DNN.

Examples of MSBPF characteristics predicted using the first DNN are shown in Fig. 11. In some cases, the characteristics can be predicted with moderate accuracy, while in other cases they show large errors.

Fig. 12 shows the validation loss when training with different dataset sizes, and it was confirmed that the validation loss decreases roughly in proportion to the power of the dataset size [35]. As mentioned in Introduction, building a highly accurate surrogate model of MSL circuits requires a large amount of training data generated by EM simulations, implying a massive computational cost.

4.3 DNN for Error Evaluation

The architecture of the second DNN for error evaluation and training settings are shown in Fig. 13 and Table 3, respectively. The convolutional layer is identical to the first DNN, while the number of nodes in the fully connected layer is changed because the output is a one-dimensional real number. The second DNN was trained to output the RMSE of the first DNN's prediction from the circuit layout, as shown in Fig.13. The dataset size is therefore equal to the number of test data for the first DNN in Table 2. The criteria for early stopping and the computer architecture are the same as for training the first DNN. The average minimum validation loss is 0.39, which is obtained through five-fold cross-validation for which it takes about 5 minutes. The average correlation coefficient between ground truth and predicted values, 0.76, is considered acceptable because the second DNN does not directly affect the accuracy of the performance prediction. Fig. 14 and Fig. 15 show one of the results of cross-validation, and they are the loss curves and prediction results on the test data, respectively.

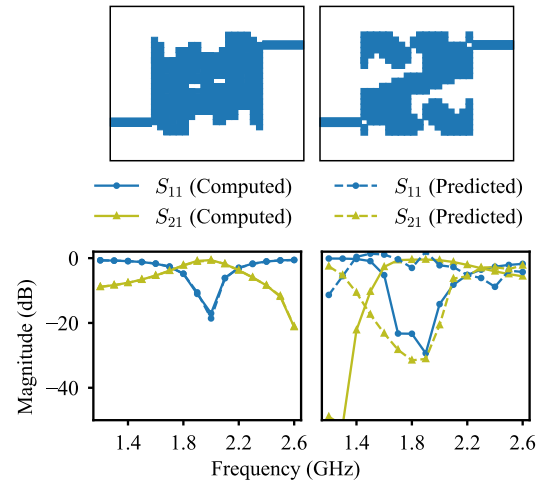


Fig. 11 Examples of predictions using the first DNN.

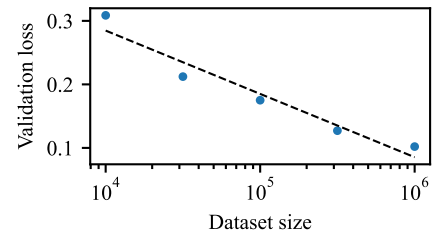


Fig. 12 Performance improvement of the first DNN by increasing the dataset size.

Table 3 Training settings for the second DNN.

Dataset size	13,395
Data split ratio (training : validation : test)	7:1:2
Update algorithm	Adam
Loss function	MSE (6)
Batch size	128
Initial learning rate	0.001
Number of epochs	117

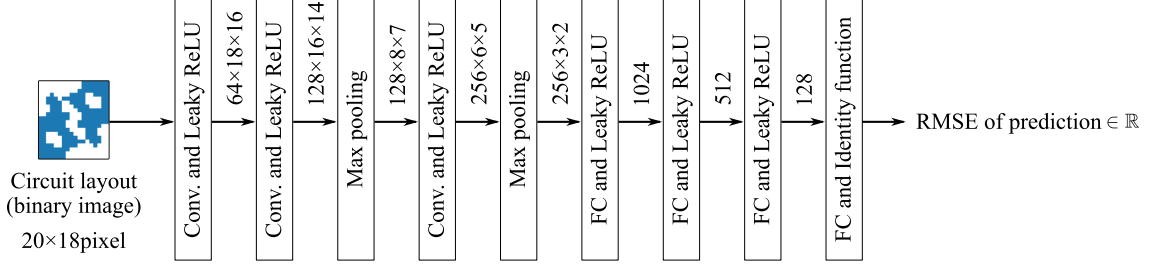


Fig. 13 Architecture of the second DNN for error evaluation. Conv. and FC represent the convolutional and fully connected layers, respectively. Angle of the negative slope for Leaky ReLU function is 0.01. The output data, RMSE of prediction, are standardized.

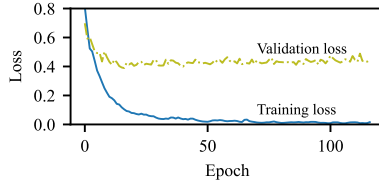


Fig. 14 Loss curves for the second DNN.

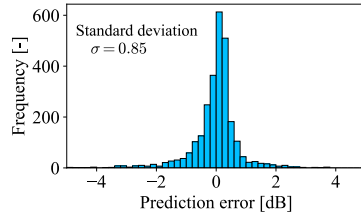


Fig. 15 Prediction results on the test data with the second DNN.

4.4 Topology Optimization

The proposed method is applied to the design of MSBPFs to verify its effectiveness. The twin DNNs mentioned in sections 4.2 and 4.3 are used. The passband settings of the MSBPFs to be designed in this section are summarized in Table 4. The passband is changed because it would make no sense to perform the optimization with the surrogate model if the settings are identical to those used in the preliminary optimization to generate the dataset. The optimization model, objective function, Gaussian basis distribution, and optimization settings are the same as in the preliminary optimization and correspond to Fig. 5, (5), Fig. 6, and Table 1, respectively.

Table 5 shows the optimization results with the four methods shown in the right part of Fig. 4: conventional methods (i) and (ii), proposed methods (i) and (ii). The threshold L_{th} is set to 2.0 dB because 95.9% of the data in the error histogram in Fig. 15 were predicted with an error of less than 2.0 dB in absolute value. The optimization is performed six times for each method with different random number seeds and the results are averaged. For methods other than the conventional method (i), F was reanalyzed by EM simulation. With regard to setting A, the conventional method (i) took about 11 hours for a single optimization. On the other hand, replacing the EM simulations with the DNN resulted in a speed-up of about 200 times (conventional method (ii)). However, as shown in

Table 4 MSBPF passband settings.

		Center frequency	Fractional bandwidth
Preliminary optimization (Section 4.1)			
		2.0 GHz	20.0%
Optimization with proposed method (Section 4.4)	A	1.6 GHz	25.0%
	B	2.1 GHz	28.6%
	C	1.9 GHz	31.6%

Table 5 Summary of optimization results.

Target filter setting (Table 4)	Optimization method	F (minimize)	Prediction error (dB)	Optimization time* (min)	Speedup ratio (-)
A	Conventional (i)	34.48	N/A	641	N/A
	Conventional (ii)	3378	8.87	3	214
	Proposed (i)	65.00	2.66	239	2.68
	Proposed (ii)	56.75	1.82	121	5.30
B	Conventional (i)	45.89	N/A	538	N/A
	Conventional (ii)	2648	7.20	3	179
	Proposed (i)	53.78	1.83	250	2.15
	Proposed (ii)	53.48	2.05	44	12.2
C	Conventional (i)	49.98	N/A	575	N/A
	Conventional (ii)	4263	9.23	3	192
	Proposed (i)	53.01	0.97	181	3.18
	Proposed (ii)	54.02	1.86	62	9.27

* "Optimization time" does not include the time required to generate the dataset and train the DNNs, which is about 24 days for all optimization methods except "Conventional (i)". The optimizations are performed on Intel® Xeon® W-2135 CPU at 3.70 GHz.

Fig. 10, the first DNN is not accurate enough, resulting in misleading and the final value of F remains significantly large. The proposed method (i) was able to reduce the prediction error to less than half, while the speed-up ratio was only about three. The proposed method (ii), which performs EM simulations on only the top 50% of individuals, is about two times faster than the proposed method (i) while maintaining the same level value of F and prediction error, and about five times faster than the conventional method (i). As mentioned above, Table 5 shows the results of reanalyzing the elite individual of the last generation with the EM simulation. It would be possible to find better solutions by selecting the best solution from the elite individuals of all generations, which are evaluated by EM simulation in the post processing. This method was found to lead to a solution with $F = 35.31$, comparable to the result of the conventional (i) ($F = 34.48$). The speed-up ratio decreases from 5.30 to 3.98 due to the increase in the EM simulations.

Fig. 16 shows the averaged convergence curves of the objective function for setting A. All the prediction-based optimizations seem to yield better F than conventional method (i), which is optimized only with EM simulations.

However, when the optimal solutions are reanalyzed with EM simulations, they are found to give worse results than conventional (i). This discrepancy is due to errors in the first DNN for S-parameter prediction. The two proposed methods are able to significantly reduce this discrepancy compared to the conventional method (ii). One of the optimization results for setting A obtained by the proposed method (ii) is shown in Fig. 17, where the measured S-parameters, together with those predicted by the DNN, and reanalyzed by EM simulation are found to be in agreement. In addition, S_{11} was less than -20 dB in the passband, indicating that the desired characteristics were obtained even though the passband was different from that of the preliminary optimization.

The results in Table 5 also show that the proposed method is also effective for B and C, and the optimizations are accelerated by an average factor of nine by the proposed method (ii). These results suggest that although the generation of training data needs computing cost, once the DNNs are trained, they can be reused to perform various optimizations at high speed.

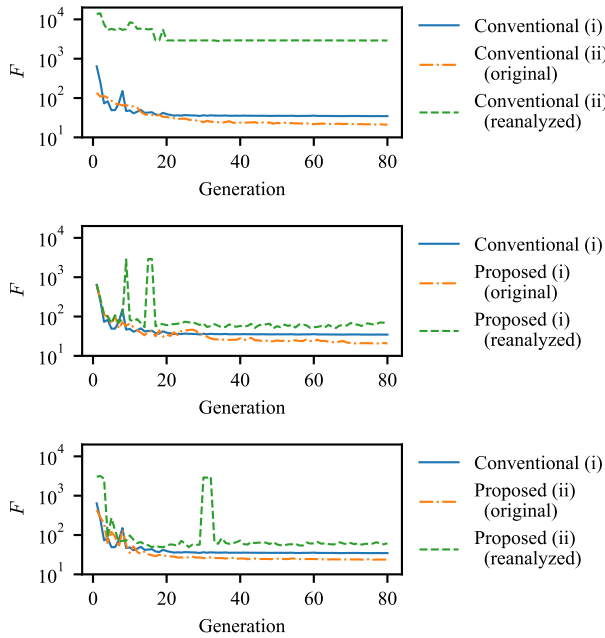


Fig. 16 Averaged convergence curves of the objective function for setting A.

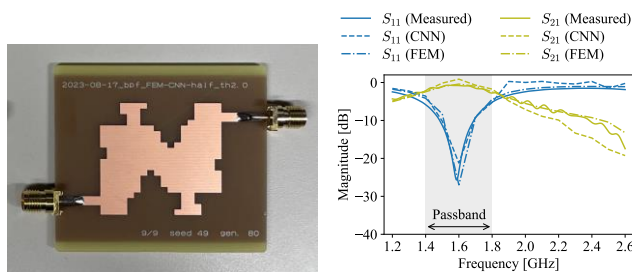


Fig. 17 One of the optimization results for setting A obtained by the proposed method (ii).

5. Conclusion

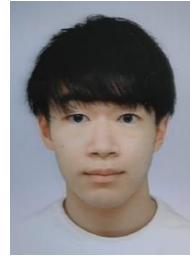
We have proposed a TO method based on the twin DNNs for prediction of S-parameters and its error evaluation. The proposed method has been applied to the optimal design of an MSBPF. It has been shown that the proposed TO works about nine times faster than that with conventional EM simulation, while avoiding misleading caused by the prediction error of the DNN. The proposed method would speed up TO using other surrogate models, even when the training data is not enough to sufficiently improve the prediction accuracy.

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